

CURRICULUM VITAE

CV SECTION 1: Employment History/Awards

NAME

Nathan Dean Clark

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EDUCATION

- 2014 Master of Agriculture in Agricultural Education,
Colorado State University, Fort Collins, CO
- 2001 Completion of Teacher Licensure Program in Agricultural Education,
University of Arizona, Tucson, AZ
- 2000 Bachelor of Science, Agriscience,
Michigan State University, East Lansing, MI

ACADEMIC POSITIONS

- 2020 to present Senior Instructor in Agricultural Education, Department of Agricultural and
Resource Economics, Colorado State University
- 2012 to 2020 Instructor in Agricultural Education, Department of Agricultural and Resource
Economics, Colorado State University

OTHER POSITIONS

- 2005 to 2012 Agriscience Instructor/FFA Advisor, Windsor High School, Windsor, CO
- 2001 to 2005 Agriscience Instructor/FFA Advisor, Millennium High School, Goodyear, AZ

CURRENT JOB DESCRIPTION

2017 to present	<u>60</u>	% Teaching	<u>40</u>	% Service/Outreach
2016-2017	<u>80</u>	% Teaching	<u>20</u>	% Service/Outreach
2015-2016	<u>90</u>	% Teaching	<u>10</u>	% Service/Outreach
2012-2015	<u>100</u>	% Teaching	<u>0</u>	% Service/Outreach

My appointment distribution has changed over the past few years as I have taken on different roles and responsibilities. After the 2014 opening of the CoBank Center for Agricultural Education, I started taking on many management roles as well as hosting, interacting, and servicing various groups at the facility. These groups include many stakeholders and donors to the College and University. As a result, my outreach appointment increased. My appointment changed again as the College increased the number of courses/students utilizing the Agricultural Research, Development, and Educational Center (ARDEC) campus as an educational space. As a result, I began serving as a facilitator of experiential learning at the ARDEC campus. As this role has evolved, the service component of my appointment has increased to more accurately reflect my academic and service/outreach responsibilities.

HONORS AND AWARDS

2024	Honorary State FFA Degree; CDE Superintendent , Colorado FFA Association (Colorado)
2017	Honorary State FFA Degree , Colorado FFA Association (Colorado)
2017	Outstanding Service Citation , Colorado Vocational Agriculture Teachers Association (Colorado)
2016	Colorado Vocational Agriculture Teachers Association 10-year service award , Colorado Vocational Agriculture Teachers Association, (Colorado)
2008	Windsor FFA Honorary Chapter FFA Degree , Windsor High School FFA Chapter (Colorado)
2005	Extra Mile Award , Windsor High School (Colorado)
2004	Golden Apple Award , Millennium High School (Arizona)
2004	Arizona Agriculture Teachers Association Ideas Unlimited 2nd runner- up , Arizona Agriculture Teachers Association (Arizona)
2002	Arizona Agriculture Teachers Association District Teacher of the Year , Arizona Agriculture Teachers Association (Arizona)
2000	Outstanding Senior Finalist , Michigan State University College of Agriculture and Natural Resources (Michigan)

CV SECTION 2: Publications/Scholarly Record

Although I do not have a research appointment, as a trained educator and as a trainer of agricultural educators, I have conducted research in the field. All my research and subsequent papers/posters focus on effective teaching, effective delivery of content, and assessment of content. I use data from this research to drive how I structure, deliver, and assess students in the classes I teach.

PUBLISHED WORKS

Refereed Journal Articles:

Bennett, J. E., Baxter, A., Enns, K. J., & **Clark, N.** (2026). Investigating the Sources and Influences Shaping Pedagogical Content Knowledge among Colorado Agricultural Educators. *Journal of Agricultural Education*, 67(1), Article 3. <https://doi.org/10.5032/jae.v67i1.3218>

Bennett, J. E., Pate, M. L., Enns, K. J., Burrows, M. S., **Clark, N.**, & Frasier, W. M. (2025). An Examination of the Effectiveness of an Institutional Intervention to Improve Retention at an Agricultural College: Effectiveness of an Institutional Intervention to Improve Retention. *NACTA Journal*, 69(1). <https://doi.org/10.56103/nactaj.v69i1.210>

Clark, N., Enns, K. J., Martin, M. Agricultural Mechanic Psychomotor Skill Development for Pre-Service Teachers: An Action Research Approach. *Journal of Agricultural Education*. *Submitted*.

Clark, N.D., Enns, K.J., Johnson, J.J. (2013). Skill-based Student Assessment: as easy as 1.2.3...4. *The Agricultural Education Magazine*. Sept/Oct Edition.

Refereed Proceedings/Transactions:

Eshelman, K., **Clark, N.**, (2024). RAM Scholars: Integrating Teacher Development in Inclusive Higher Education, In research poster proceedings of the Western Region of American Association of Agricultural Education research conference, Chico, CA.

Bennett, J. E., Enns, K. J., **Clark, N.**, Baxter, A., (2023). Examining a Perception-based Pedagogical Content Knowledge Survey for Agricultural Educators in Colorado, In research poster proceedings of the Western Region of American Association of Agricultural Education research conference, Logan, UT.

Bennett, J. E., Enns, K. J., **Clark, N.**, (2023). Using Agricultural Literacy to develop a Community of Practice for SBAE Students. In poster proceedings of the of American Association for Agricultural Education research conference, Raleigh, NC.

Bennett, J. E., Enns, K. J., **Clark, N.**, Martin, M. (2020). Teaching and Learning Scholarship. In poster proceedings of the Western Region of American Association of Agricultural Education research conference, Virtual Conference.

Clark, N., Enns, K. J., Martin, M. (2019). Crate it Away! Construction of a Wooden Crate in a Power Tool Skill Development Course. In poster proceedings of the Western Region of American Association of Agricultural Education research conference, Anchorage, AK.

Enns, K.J., **Clark, N.**, Bennett, J. E., (2019). Building ALL Three-legs to the Teacher Retention Stool. In poster proceedings of the Western Region of American Association of Agricultural

Education research conference, Anchorage, AK.

Frasier, W. M., **Clark, N.**, Martin, M., (2018). Experiential Learning for Freshman in Agriculture. In poster proceedings of the Western Region of American Association of Agricultural Education research conference, Boise, ID.

Clark, N., Enns, K. J. (2015). Impacts of a Skill Development Course on Teacher Candidate Confidence in their Knowledge, Skill, Experience, and Teaching of Common Power Tools. San Antonio, TX.

Clark, N., Enns, K. J., Martin, M. (2014). Agricultural Mechanic Psychomotor Skill Development for Pre-Service Teachers: An Action Research Approach. Research paper proceedings of the of Western Region of American Association for Agricultural Education research conference, Kona, HI. (pp. 149-162).

Enns, K. J., **Clark, N.**, Martin, M. (2014). Becoming Professionals: Utilizing Ideas Unlimited and Professional Development Seminars to Foster Success in Future Teachers. In poster proceedings for American Association for Agricultural Education Research Conference, Snowbird, UT.

Clark, N.D., & Enns, K.J., (2013). Agricultural Mechanics Round Robin Training Seminar. Poster Proceedings for American Association for Agricultural Education Research Conference, Columbus, OH. (pp. 11-14).

CV SECTION 3: EVIDENCE OF TEACHING AND ADVISING EFFECTIVENESS

As an instructor, my primary appointment is to teach undergraduate courses and to supervise student teaching experiences in the Agricultural Education major. However, in addition to teaching courses, I also facilitate/coordinate the College of Agricultural Sciences experiential learning initiatives at the Agricultural Research, Development, and Educational Center (ARDEC). This campus is considered CSU's "flagship" experiment station research farm and is located about 12 miles from main campus. This initiative is designed to provide students in the College of Agricultural Sciences with opportunities to not only hear about agriculture production and research, but to experience it as well. I work with faculty who teach courses at the ARDEC campus, helping them develop experiential learning opportunities in their classes both on the ARDEC farm and at surrounding farms, agricultural business, research centers, etc. In addition to helping faculty, I also manage and maintain the teaching spaces at ARDEC. As we are not on main campus, I help to coordinate the scheduling of courses, transportation of students, technology needs of faculty, etc. I also developed and designed an innovative teaching classroom in the ARDEC Taylor Auditorium. This auction ring auditorium now seats 200 students at tables and chairs in groups with seven monitors around the room. I developed the technology to connect all the monitors together both hardwired and most recently wirelessly. In addition to the Taylor Auditorium, I manage the instructional needs of instructors in the CoBank Center for Agricultural Education. This includes the management of four classrooms as well as a large mechanical and technical teaching laboratory. Through this facilitation of learning, I have an impact on several students outside the courses I teach. Below is a summary of the courses I teach/have taught as an instructor at CSU.

AGED 110: Agriculture Production Systems

3 credit hours (2 credits lecture, 1 credit 3 clock hour lab)

Catalog Description: Broad survey of the diverse aspects of Colorado agriculture.

This is a course I developed to address the needs of students in the College who want to know more about the diversity of production agriculture in Colorado. This is a field experience course in which we spend time participating in various production systems and reflect on how these experiences are related to their everyday lives as students of agriculture, consumers, and producers.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025**	Spring	AGED 110 Agriculture Production Systems	2	10	20
2025	Spring	AGED 110 Agriculture Production System Lab	1	10	10
2024***	Summer	AGED 110 Agriculture Production Systems-Online	3	12	36
2024**	Spring	AGED 110 Agriculture Production Systems	2	11	22
2024	Spring	AGED 110 Agriculture Production System Lab	1	11	11
2023**	Spring	AGED 110 Agriculture Production Systems	2	11	22
2023	Spring	AGED 110 Agriculture Production System Lab	1	11	11
2022	Spring	AGED 110 Agriculture Production Systems	2	9	18
2022	Spring	AGED 110 Agriculture Production System Lab	1	9	9
2021	Spring	AGED 110 Agriculture Production Systems	2	10	20
2021	Spring	AGED 110 Agriculture Production System Lab	1	10	10
2020	Spring	AGED 110 Agriculture Production Systems	2	6	12
2020	Spring	AGED 110 Agriculture Production System Lab	1	6	6
2019	Spring	AGED 110 Agriculture Production Systems	2	14	28
2019	Spring	AGED 110 Agriculture Production System Lab	1	14	14
2018	Spring	AGED 110 Agriculture Production Systems	2	8	16
2018	Spring	AGED 110 Agriculture Production System Lab	1	8	8
2016	Fall	AGED 110 Agriculture Production Systems	2	17	34

2016	Fall	AGED 110 Agriculture Production System Lab	1	17	17
2015	Fall	AGED 110 Agriculture Production Systems	2	16	32
2015	Fall	AGED 110 Agriculture Production System Lab	1	16	16
2014	Fall	AGED 110 Agriculture Production Systems	2	11	22
2014	Fall	AGED 110 Agriculture Production System Lab	1	11	11
2013	Fall	AGED 280A3 Agriculture Production System*	1	8	8
2013	Fall	AGED 280A3 Agriculture Production System Lab*	1	8	8

*This course was first offered with an experimental number for only 2 credits. When the course was made permanent, I added one credit to the course.

** In Spring 2023 the first Ram Scholars student enrolled in this course. The Ram Scholars Program is a pioneering initiative aimed at offering college education in agriculture for individuals with intellectual and developmental disabilities, I played a pivotal role in shaping and delivering the first integrated class for students in this groundbreaking program. Tasked with developing curriculum content tailored to the unique needs and learning styles of the students, I collaborated closely with the first Ram Scholar student and the peer mentors to ensure a supportive and enriching educational experience. This experience allowed me to demonstrate adaptability, creativity, and a commitment to inclusive education. Through innovative teaching methods and personalized approaches, I fostered a positive learning environment that empowered students to excel in their academic pursuits. Being part of the initiation of the Ram Scholars Program reflects my dedication to inclusive education and a passion for creating opportunities for all individuals to access quality higher education in the field of agriculture. In the Spring of 2024, I continued to work with the Ram Scholars program and 6 of the 11 students enrolled in the course were neurodiverse learners full integrated into this course.

***In summer 2024 I developed an Online version of this course for 4-H members to earn college credit for a participating in 4-H. The students enrolled in the course, finished their 4-H projects in their local communities and earned credit for completing this course. Students also completed projects similar to what traditional RI students complete. This course creates an opportunity for 4-H members to consider CSU as a higher education option as they have already completed a course before graduation.

AGED 220: Understanding Agricultural Education

1 credit hour (1 credit lecture)

Catalog Description: Understanding different agricultural education systems. Understanding delivery models of agricultural education programs.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2024	Fall	AGED 220 Understanding Agricultural Education	1	24	24
2023	Fall	AGED 220 Understanding Agricultural Education	1	16	16
2022	Fall	AGED 220 Understanding Agricultural Education	1	24	24
2021	Fall	AGED 220 Understanding Agricultural Education	1	22	22
2020	Fall	AGED 220 Understanding Agricultural Education	1	22	22

AGED 240: Technical Tool Applications in Ag Education

2 credit hours (1 credit lecture, 1 credit - 3 clock hour lab)

Catalog Description: Development of safe competencies and applications related to power and technical tools utilized in school-based agricultural education programs.

I developed this course to address the needs of Ag Ed students training to become agriculture teachers. As teachers, they are expected to teach courses in the agriculture mechanical pathway and have students operate several technical and power tools. In the course, students learn how to safely operate and how to teach the

safe operation of common technical tools found in ag mechanics teaching labs.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Spring	AGED 240 Technical Tool Applications in Ag Education	1	10	10
2025	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	10	10
2024	Spring	AGED 240 Technical Tool Applications in Ag Education	1	5	5
2024	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	5	5
2023	Spring	AGED 240 Technical Tool Applications in Ag Education	1	5	5
2023	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	5	5
2022	Spring	AGED 240 Technical Tool Applications in Ag Education	1	9	9
2022	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	9	9
2021	Spring	AGED 240 Technical Tool Applications in Ag Education	1	9	9
2021	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	9	9
2020	Spring	AGED 240 Technical Tool Applications in Ag Education	1	3	3
2020	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	3	3
2019	Spring	AGED 240 Technical Tool Applications in Ag Education	1	8	8
2019	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	8	8
2018	Spring	AGED 240 Technical Tool Applications in Ag Education	1	13	13
2018	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	13	13
2017	Spring	AGED 240 Technical Tool Applications in Ag Education	1	4	4
2017	Spring	AGED 240 Technical Tool Applications in Ag Education Lab	1	4	4
2016	Fall	AGED 240 Technical Tool Applications in Ag Education	1	8	8
2016	Fall	AGED 240 Technical Tool Applications in Ag Education Lab	1	8	8
2015	Fall	AGED 240 Technical Tool Applications in Ag Education	1	20	20
2015	Fall	AGED 240 Technical Tool Applications in Ag Education Lab	1	20	20
2014	Fall	AGED 240 Technical Tool Applications in Ag Education	1	8	8
2014	Fall	AGED 240 Technical Tool Applications in Ag Education Lab	1	8	8
2014	Spring	AGED 240 Technical Tool Applications in Ag Education	1	6	6
2013	Fall	AGED 240 Technical Tool Applications in Ag Education	1	13	13

AGED 241: Plumbing and Electrical Applications in Ag Ed

1 credit hour (1 credit lecture)

Catalog Description: Development of competencies and theory related to plumbing and electrical applications utilized in school-based agricultural education programs.

I developed this course to address the needs of student entering the Ag Ed profession who are expected to teach content in the agriculture mechanical pathway. However, in 2017 this content was merged with the AGED 244 class and was then discontinued.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2017	Spring	AGED 241 Plumbing and Electrical Applications in Ag Ed	1	7	7
2016	Spring	AGED 241 Plumbing and Electrical Applications in Ag Ed	1	18	18
2015	Spring	AGED 241 Plumbing and Electrical Applications in Ag Ed	1	11	11
2013	Fall	AGED 241 Plumbing and Electrical Applications in Ag Ed	1	13	13

AGED 244: Power, Structure and Technical Systems in Ag Ed

3 credit hours (2 credits lecture, 1 credit-3 clock hour lab)

Catalog Description: Development of competencies and theory related to agricultural power, structure, and technical systems utilized in school-based agricultural education programs.

This is a course I developed to address the needs of Ag Ed students training to become agriculture teachers. As teachers, they are expected to teach courses in the agriculture mechanical pathway but had no coursework in this content area and often no confidence to teach in this space. This course provides material and competency for future teachers.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	15	18
2025	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	15	9
2024	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	9	18
2024	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	9	9
2023	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	3	6
2023	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	3	1
2022	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	8	16
2022	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	8	8
2021	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	4	8
2021	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	4	4
2020	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	5	10
2020	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	5	5
2019	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	4	8
2019	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	4	4
2018	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	10	20
2018	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	10	10
2017	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed	2	8	16
2017	Fall	AGED 244 Power, Structure and Technical Systems in Ag Ed Lab	1	8	8
2017	Spring	AGED 244 Power Systems in Agricultural Education*	1	6	6
2016	Spring	AGED 244 Power Systems in Agricultural Education*	1	16	16
2015	Spring	AGED 244 Power Systems in Agricultural Education*	1	10	10
2014	Spring	AGED 244 Power Systems in Agricultural Education*	1	7	7

*The first offerings were a 1 credit lecture course taught in conjunction with a separate laboratory course. In 2017 we revised this course, merged the content from another 1 credit course and added a lab moving it from 1 credit to 3 credits.

AGED 320: Technology Lab for Ag Education

1 credit hour (1 credit-3 clock hour lab)

Catalog Description: Laboratory applications related to the power, structure, and technical systems pathway utilized in school-based agricultural education programs.

This course is designed for students who are preparing to student teach to hone skills they feel they need more confidence to teach. They receive individualized instruction based on self-identified deficiencies in their skill set.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2024*	Fall	AGED 320 Technology Lab for Ag Education Lab	1	5	5
2024	Fall	AGED 320 Technology Lab for Ag Education Lab	1	2	2
2024	Spring	AGED 320 Technology Lab for Ag Education Lab	1	2	2
2023*	Fall	AGED 320 Technology Lab for Ag Education Lab	1	7	7
2023	Fall	AGED 320 Technology Lab for Ag Education Lab	1	2	2

2022	Fall	AGED 320 Technology Lab for Ag Education Lab	1	3	3
2022	Spring	AGED 320 Technology Lab for Ag Education Lab	1	1	1
2021	Fall	AGED 320 Technology Lab for Ag Education Lab	1	1	1
2021	Spring	AGED 320 Technology Lab for Ag Education Lab	1	2	2
2020	Fall	AGED 320 Technology Lab for Ag Education Lab	1	2	2
2020	Spring	AGED 320 Technology Lab for Ag Education Lab	1	2	2
2019	Fall	AGED 320 Technology Lab for Ag Education Lab	1	2	2
2018	Fall	AGED 320 Technology Lab for Ag Education Lab	1	1	1
2017	Fall	AGED 320 Technology Lab for Ag Education Lab	1	6	6
2017	Spring	AGED 320 Technology Lab for Ag Education Lab	1	7	7
2016	Fall	AGED 320 Technology Lab for Ag Education Lab	1	4	4
2016	Spring	AGED 320 Technology Lab for Ag Education Lab	1	16	16
2015	Fall	AGED 320 Technology Lab for Ag Education Lab	1	4	4
2015	Spring	AGED 320 Technology Lab for Ag Education Lab	1	10	10
2014	Spring	AGED 320 Technology Lab for Ag Education Lab	1	9	9
2013	Fall	AGED 320 Technology Lab for Ag Education Lab	1	14	14

*I created a special section of AGED 320 in the fall of 2023 and 2024 to complement my work in the pilot Ram Scholars Program. Ram Scholars is a pioneering initiative, dedicated to providing college education in agriculture to individuals with intellectual and developmental disabilities. Within this course, students were immersed in experiential learning opportunities in the realm of production agriculture. Notably, each Ram Scholar student was paired with peer mentors drawn from various disciplines, including Agriculture Education, Human Development and Family Studies, and Occupational Therapy. In addition to teaching, my responsibilities extended to designing and implementing experiential labs, ensuring that the learning environment was both enriching and inclusive. Furthermore, I took charge of training and managing the peer mentors, fostering a supportive framework for the participants. This experience underscores my commitment to inclusive education and showcases my ability to design and execute specialized curriculum components, manage experiential learning initiatives, and oversee peer mentoring programs. It reflects my dedication to fostering educational opportunities for individuals with diverse learning needs in the field of agriculture.

AGED 420: Developing School-Based Ag Education Programs

3 credit hours (3 credits lecture)

Catalog Description: Developing knowledge in the approach and delivery of school-based agricultural education programs.

I taught this course for one semester for a colleague who was on a sabbatical.

2023: I added this course to my teaching load in the Spring of 2023 to increase my work with the Teacher Development Ag Ed Students

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025**	Spring	AGED 420 Developing School-Based Ag Education Programs	3	2	6
2025*	Spring	AGED 420 Developing School-Based Ag Education Programs	3	8	24
2024	Spring	AGED 420 Developing School-Based Ag Education Programs	3	3	9
2023	Spring	AGED 420 Developing School-Based Ag Education Programs	3	4	12
2016	Spring	AGED 420 Developing School-Based Ag Education Programs	3	8	24

*This course was co-taught during the spring. My contribution to this course was 30%

** I taught this course in the spring of 2025 as a special section for the Ag Ed distance education NE3 program. This content was delivered in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO. For more information on the NE3 program see description in Development of New Courses/Programs section below.

AGED 440: Managing Experiences in Ag Ed Laboratories

1 credit hour (1 credit-3 clock hour lab)

Catalog Description: Theory, management, and pedagogy of delivering safety instruction and experiential curriculum in secondary agricultural education laboratory settings.

This is a new course I developed to help students learn how to effectively plan, deliver, and assess content in experiential learning situations. Students learn steps in preparing to teach labs, field trips, group projects, etc. and how to assess student learning in the process.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	8	4
2024	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	4	4
2023	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	4	4
2022	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	4	4
2021	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	2	2
2020	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	3	3
2019	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	6	6
2018	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	9	9
2017	Fall	AGED 440 Managing Experiences in Ag Ed Laboratories Lab	1	13	13

AREC 484: Supervised College Teaching

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2017	Fall	AREC 484 Supervised College Teaching	2	2	4

AGED 495: Independent Study

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2016	Spring	AGED 495 Independent Study	3	3	9

AGED 496: Group Study

This course was offered for the Ram Scholars summer internship in production agriculture. This is a pilot program designed for individuals with intellectual and developmental disabilities (IDD). The program involves a classes and internship where Ram Scholars are paired with peer mentors from both the Agriculture Education and Human Development and Family Studies Departments.

During the internship, peer mentors work alongside Ram Scholars to provide hands-on training in production agriculture practices. The goal is to empower and educate individuals with IDD in the field of agriculture, allowing them to gain valuable skills and experience. Through this class I oversaw and coordinated the overall internship experience. This involved training and supporting peer mentors, establishing connections with the agricultural industry, and facilitating the learning process for both Ram Scholars and their mentors. The collaborative nature of the program, with mentors from diverse backgrounds, reflects a holistic approach to education and skill-building for individuals with intellectual and developmental disabilities in the context of production agriculture. This initiative not only provides practical knowledge but also emphasizes the importance of inclusive education and community engagement in the agricultural field.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2023	Summer	AGED 496 Group Study-Ram Scholars Summer Internship Pilot	1	6	6

AGED 540: Ag Ed Laboratory Management and Safety

2 credit hours (2 credits lecture)

Catalog Description: Theory, management, and pedagogy of delivering safety instruction and experiential curriculum in secondary agricultural education laboratory settings.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2024	Fall	AGED 540: Ag Ed Laboratory Management and Safety	2	1	2
2024	Summer	AGED 540: Ag Ed Laboratory Management and Safety	2	3	6
2021	Fall	AGED 540: Ag Ed Laboratory Management and Safety	2	1	2
2021	Summer	AGED 540: Ag Ed Laboratory Management and Safety	2	2	4
2020	Summer	AGED 540: Ag Ed Laboratory Management and Safety	2	2	4

AGED 587: Internship in Extension

1 to 2 credit hours (Internship)

Catalog Description: First-hand experiences in extension programming.

I developed this course to fit into the redesigned Extension Education Master's degree our Ag Ed program took over recently.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2018	Summer	AGED 587 Internship in Extension	2	2	4

EDCT 492: Seminar-Professional Relations

1 credit hour (1 credit seminar)

Catalog Description: Collegial and professional discussions, support, and assistance.

This course is the seminar for students who are in the field completing their student teaching experience. I coordinate these seminars during the semester to bring the student teachers back to campus and evaluate their experiences, give them advice, etc.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Spring	EDCT 492 Seminar-Professional Relations	1	4	4
2024	Fall	EDCT 492 Seminar-Professional Relations	1	2	2
2024	Spring	EDCT 492 Seminar-Professional Relations	1	2	2
2023	Fall	EDCT 492 Seminar-Professional Relations	1	2	2
2023	Spring	EDCT 492 Seminar-Professional Relations	1	3	3
2022	Spring	EDCT 492 Seminar-Professional Relations	1	2	2
2021	Fall	EDCT 492 Seminar-Professional Relations	1	1	1
2021	Spring	EDCT 492 Seminar-Professional Relations	1	2	2
2020	Fall	EDCT 492 Seminar-Professional Relations	1	2	2
2020	Spring	EDCT 492 Seminar-Professional Relations	2	5	10
2019	Spring	EDCT 492 Seminar-Professional Relations	2	6	12
2018	Spring	EDCT 492 Seminar-Professional Relations	2	7	14
2017	Spring	EDCT 492 Seminar-Professional Relations	2	3	6
2016	Spring	EDCT 492 Seminar-Professional Relations	2	6	12
2015	Spring	EDCT 492 Seminar-Professional Relations	2	9	18
2013	Fall	EDCT 492 Seminar-Professional Relations	2	14	28

EDCT 590: Workshop: Teacher Leadership in Agricultural Education

1 to 6 credit hours (Workshop)

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
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2024	Summer	EDCT 590: Workshop: CASE Curriculum PD	1 to 6	5	15
2024	Summer	EDCT 590: Workshop: Teacher Leadership in Ag Ed	1 to 6	26	90
2023	Summer	EDCT 590: Workshop: CASE Curriculum PD	1 to 6	9	27
2023	Summer	EDCT 590: Workshop: Teacher Leadership in Ag Ed	1 to 6	17	52
2022	Summer	EDCT 590: Workshop: Teaching Animal Science	1 to 6	15	45
2022	Summer	EDCT 590: Workshop: Teacher Leadership in Ag Ed	1 to 6	27	90
2021	Summer	EDCT 590: Workshop: Teacher Leadership in Ag Ed	1 to 6	21	64

AGRI 181A1: Contemporary Agricultural Systems – Co-taught (50%)

1 credit hour (1 credit-3 clock hour lab)

Catalog Description: Understanding and appreciating the breadth and interconnectedness of the diverse elements of modern agriculture.

I helped create this service course for first year students in the College of Agricultural Sciences. This course has a heavy field experience component providing students with experiences to deepen their understanding on the complexities of the production agriculture system.

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2018	Fall	AGRI 181A1 Contemporary Agricultural Systems Lab	1	40	40
2017	Fall	AGRI 181A1 Contemporary Agricultural Systems Lab	1	20	20

The following courses were developed for a new modified 3+1 program in Ag Ed called NE3. The content for each course was delivered in a hybrid distance format in northeastern Colorado. These courses are designed for students on the path to ag teacher licensure living in and around Sterling, CO. The courses are both upper division content in the six different Ag Ed pathways and courses related teacher licensure methods. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

AGED 380A1: Teaching Animal Science in School Based Ag Ed

3 Credit Hours (1 credit lecture, 1 credit recitation, 1 credit-3 clock hour lab)

Catalog Description: Students will learn in experiential form to implement secondary Ag Ed Standards through contextual learning and placements in industry and classrooms related to Animal Science.

I helped create this course in the summer of 2024 as a content course for the new Ag Ed NE3 program. I taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to gain upper division content in Animal Science. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2024	Fall	AGED 380A1: Teaching Animal Science in School Based Ag Ed	3	2	6

AGED 380A2: Teaching Nat Res/Env Sc in School Based Ag Ed

3 Credit Hours (1 credit lecture, 1 credit recitation, 1 credit-3 clock hour lab)

Catalog Description: Learn in experiential form to implement secondary Ag Ed Standards through contextual learning and placements in industry and classrooms related to Natural Resources/Environmental Science.

I helped create this course in the summer of 2024 as a content course for the new Ag Ed NE3 program. I

taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to gain upper division content in Natural Resources and Environmental Science. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2024	Fall	AGED 380A2: Teaching Nat Res/Env Sc in School Based Ag Ed	3	2	6

AGED 380A3: Teaching PSTS in School Based Ag Ed

3 Credit Hours (1 credit lecture, 1 credit recitation, 1 credit-3 clock hour lab)

Catalog Description: Students will learn in experiential form to implement secondary Ag Ed Standards through contextual learning and placements in industry and classrooms related to Power, Structural, & Technical Systems.

I helped create this course in the summer of 2024 as a content course for the new Ag Ed NE3 program. I taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to gain upper division content in Power, Structure and Technology. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2024	Fall	AGED 380A3: Teaching PSTS in School Based Ag Ed	3	2	6

AGED 380A4: Teaching Plant Science in School Based Ag Ed

3 Credit Hours (1 credit lecture, 1 credit recitation, 1 credit-3 clock hour lab)

Catalog Description: Students will learn in experiential form to implement secondary Ag Ed Standards through contextual learning and placements in industry and classrooms related to Animal Science.

I helped create this course in the fall of 2024 as a content course for the new Ag Ed NE3 program. I taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to gain upper division content in Plant Science. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Spring	AGED 380A4: Teaching Plant Science in School Based Ag Ed	3	2	6

AGED 380A5: Teaching Food Products and Processing Systems in School Based Ag Ed

3 Credit Hours (1 credit lecture, 1 credit recitation, 1 credit-3 clock hour lab)

Catalog Description: Students will learn in experiential form to implement secondary Ag Ed Standards through contextual learning and placements in industry and classrooms related to the Food Products and Processing pathway.

I helped create this course in the fall of 2024 as a content course for the new Ag Ed NE3 program. I taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to gain upper division content in the Food Products and Processing Systems pathway. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Spring	AGED 380A5: Teaching Food Products and Processing in SBAE	3	2	6

AGED 380A6: Teaching Agribusiness Systems in School Based Ag Ed

3 Credit Hours (1 credit lecture, 1 credit recitation, 1 credit-3 clock hour lab)

Catalog Description: Students will learn in experiential form to implement secondary Ag Ed Standards through contextual learning and placements in industry and classrooms related to the Agribusiness pathway.

I helped create this course in the fall of 2024 as a content course for the new Ag Ed NE3 program. I taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to gain upper division content in the Agribusiness Systems pathway. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Spring	AGED 380A6: Agribusiness Systems in SBAE	3	2	6

EDUC 380A4: Technology and Assessment in Ag Ed

2 Credit Hours (1 credit lecture, 1 credit-2 clock hour lab)

Catalog Description: Agriculture educators develop key skills and strategies for the use of educational technology and assessment tools in educational and classroom settings.

I helped create this course in the summer of 2024 as a content course for the new Ag Ed NE3 program. I taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to take education courses required for completing the teacher licensure program. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2024	Fall	EDUC 380A4: Technology and Assessment in Ag Ed	3	2	6

EDUC 380A5: Instr I-Individualization/Management in Ag Ed

3 Credit Hours (3 credit lecture)

Catalog Description: Theory, research, and practice of teaching at the junior high/middle school level; adapting instruction for individuals including learners with special needs in agricultural education.

I helped create this course in the fall of 2024 as a content course for the new Ag Ed NE3 program. I taught this course in a hybrid distance format in northeastern Colorado. This course is designed for students on the path to ag teacher licensure living in and around Sterling, CO to take education courses required for completing the teacher licensure program. *For more information on the NE3 program see description in Development of New Courses/Programs section below.*

<u>Year</u>	<u>Semester</u>	<u>Course No./Title</u>	<u>Cr. Hrs.</u>	<u>Enrollment</u>	<u>SCH</u>
2025	Spring	EDUC 380A5: Instr I-Individualization/Management in Ag Ed	3	2	6

Course Syllabi, Assignments, and Other Materials

Below are two descriptions of courses I teach and reasons why I included syllabi of them in the appendices (available upon request).

Fall 2018, AGED 244–Power, Structure, and Tech. Systems in Ag Ed

This course is a marathon of topics taught at a sprinters pace. You can see from the diversity of topics that students are expected to learn several different topics of information in a short amount of time. They

do well in the course even though there is a significant amount of information covered.

Fall 2015, AGED 110–Agriculture Production Systems

I provided this sample because for this course I spent time this semester developing the course and working out better course assessments. The course had a better overall student survey score than the previous year. I also worked on the course assessments more and was clearer with the students about what the expectations of the lab and lecture should look like.

Student Course Surveys

In the appendices (available upon request) I have included samples from past course student surveys. As a summary, I score well in areas related to my knowledge of my content, creating a comfortable and inviting classroom environment, and increasing student's knowledge and comfort in my subject areas. It is worthy to note that over the years.

I do not always score as well as I would like in the area of providing efficient and timely feedback to students. However, this is an area I have worked to improve over time. As a result, though I still do not score as high as I would like from students, I have been able to grow and improve in this area. This is evident in survey results but also in my day-to-day interactions with students.

Development of New Courses/Programs

Below is a summary of the new courses and programs I have developed/implemented while working at CSU.

Ram Scholars Inclusive Post Secondary Education Program:

Ram Scholars is a pioneering initiative, dedicated to providing college education in agriculture to individuals with intellectual and developmental disabilities (IDD). In this program, students are immersed in experiential learning opportunities in the realm of production agriculture. Notably, each Ram Scholar student is paired with peer mentors drawn from various disciplines, including Agriculture Education, Human Development and Family Studies, and Occupational Therapy. In addition to teaching/integrating IDD students in my classes, my responsibilities extended to designing and implementing experiential labs, ensuring that the learning environment was both enriching and inclusive. Furthermore, I took charge of training and managing the peer mentors, fostering a supportive framework for the participants. The Ag Ed peers have gained practical knowledge and skills for working with diverse populations in their future classrooms. I work directly with the peers helping them reflect and learn from their peering experience and discussing how this experience can be directly translated into their future teaching. This experience underscores my commitment to inclusive education and showcases my ability to design and execute specialized curriculum components, manage experiential learning initiatives, and oversee peer mentoring programs. It reflects my dedication to fostering educational opportunities for individuals with diverse learning needs in the field of agriculture.

Northeastern Third Year (NE3) Agricultural Education Program:

The NE3 Program is a partnership program between Colorado State University (CSU) and Northeastern Junior College (NJC) whereas a transfer student from NJC studying Ag Ed at CSU will stay in northeastern Colorado during their Junior (third) year. Students in this program are CSU students but are provided with services (dorms, meal plan, library access, etc.) from NJC. During the program, students take ag content classes which are taught through local ag industry and school-based ag ed classroom placements. The students spend time immersed in ag related industries and then subsequent related classrooms to gain invaluable experiences and knowledge related to the 6 different ag ed pathways. Additionally, I travel to Sterling, CO every Friday and teach a series of 4 to 5 different classes which are designed to help the students reflect and recognize the ag content they are learning by participating in these experiences. In addition to the content reflection course, while there I also teach courses on methods of teaching, assessment, technology, etc. At the conclusion of their third year, students will come to CSU main campus for the fall semester (housing provided) to finish their methods and education course work and integrate

with the rest of their student teaching cohort. During the spring of the fourth year, the NE3 students, are on track to student teach with their cohort. Fall 2024 was the pilot year for this program and I along with a colleague have been solely responsible for the implementation of the NE3 program. This program hinges on dedication to think outside the “normal” classroom experience, the fostering of relationships with NJC faculty, local school-based Ag Ed teachers, local ag industry professionals, etc. These are all skills I have demonstrated while helping to implement this program which took less than a year to go from “idea” to enrolling and teaching students in the program.

AGED 110: Agriculture Production Systems:

This course was designed to provide students experiences in many aspects of production agriculture. Students participate in firsthand experiences related to production agriculture and relate these experiences to content they have seen in other courses, in their everyday lives, or what they have seen on the internet. This course was created to be an elective option but is included as a requirement for students whose major is Agricultural Education-Ag Literacy.

AGED 240: Technical Tool Applications in Ag Education;

AGED 241: Plumbing and Electrical Applications in Ag Ed;

AGED 244: Power, Structure and Technical Systems in Ag Ed;

AGED 320: Technology Lab for Ag Education;

These courses were all developed to address a content need for students majoring in Agricultural Education-Teacher Development. As these students are training to become secondary agriculture educators, they are expected to teach content from several different pathways. One pathway missing in the training program was in the area of agricultural mechanics. These courses were developed to address this need so students graduating from the program have the confidence to teach content in this pathway. All of these courses are required courses in the Ag Ed teacher development course of study.

AGED 440: Managing Experiences in Ag Ed Laboratories

This course was developed to address another need of the agricultural education teacher development students. From previous classes students were able to develop skills, content, and confidence to teach in the ag mechanics pathway. However, there was a need to more deeply cover content related to the management and pedagogy of teaching in laboratory settings.

AGED 587: Internship in Extension

I was responsible for the development of this course as one of several the Ag Ed program took on when the Masters Extension Education degree was taken on by the program. In the course students complete an internship in and extension program, develop, deliver, and evaluate an educational program during the internship and experience/reflect on the day-to-day life of working in Extension.

AGRI 181A1 Co-Developed: Contemporary Agricultural Systems

I was part of a team that developed this course for first year students in the College of Agricultural Sciences. The course is designed to allow students opportunities to experience various agricultural systems in Colorado and to develop a sense of community and belonging in the college. My main contribution to the course development stemmed from my close relationships with the local agriculture community. I helped to design and facilitate many of the agriculture experiences for the course.

Development of New Teaching Techniques

Teaching in the mechanical agriculture pathway led me to do research on more effective and efficient modes of delivering content to students in psychomotor skill development. Through research in the AGED 240 Technical Tools course, I developed the Agricultural Mechanics Psychomotor Skill Model. This model is designed to guide instructors in skill-based content delivery with a flipped classroom approach for delivery of content with class time dedicated to assessment and to practicing practical skill development. This model is provided below:

Agricultural Mechanics Psychomotor Skill Development Model

Students Does	Instructor Does
• Prior to class: – Read safety rules of tools being covered in class (usually 2-3 tools). – Watch a video demonstration on operation and maintenance of each tool.	• Prior to class: • Preparation of materials for students (minimum of 5 days prior to class): – Provide students with link/access to videos of tools which are to be covered (usually 2-3 tools). – Provide students with safety rules for each tool. • Preparation of class/laboratory (prior to start of class on day of): – Have safety tests ready for students to take (preferably online versions for immediate feedback). – Set up laboratory with practice stations for tools covered. – Ensure there are clear sight lines for all practice stations.
• In classroom: – Ask clarification questions from videos. – Take a safety test over each tool. – Grade and obtain a score of 100% on each test. – If a perfect score is not obtained, review videos for clarification. – After passing, fill out the safety section of the skill card	• In classroom: – Answer clarification questions and reinforce any pertinent information. Allocation of time for this task restricted to no more than 5 minutes. – Administer safety tests and grade when students are done. – Verify and sign off on skill card affirming skill level attainment of the student

• In laboratory: – Follow steps of the Job Operation Sheet to practice using tool. – Upon completion, report to the instructor with JOS for assessment of steps and feedback. – After completing the JOS, fill out more sections of the skill card indicating level of skill development	• In laboratory: – Hand out skill Job Operation Sheets and break class into rotation groups. – Station yourself with clear sight lines to all stations – Observe students practicing tool skills. – Assess student performance for each step of the JOS according to what you observed. – Provide feedback on where the student needs to improve. – Verify and sign off on skill card once more affirming skill level attainment of the student
• Future skill development – After multiple uses of tools or at the conclusion of a project student revisits the skill cards each time to indicate skill level for each tool and to show growth while enrolled in the program	• Future skill development – Verify and sign off on skill card each time students fill them out affirming the skill level attainment of each student while enrolled in the program

Other Evidence

As facilitator of experiential learning at the ARDEC campus, I have the opportunity to influence students and courses beyond my own teaching load. Since the fall of 2016, approximately 175 students are bussed on Tuesdays and Thursdays to the ARDEC campus for a cohort day of experiential learning. In addition to these students, other courses in the College of Agricultural Sciences are delivered in either the Taylor Auditorium or the CoBank Center for Agricultural Education. My role of experiential learning coordinator affords me the opportunity to work with the instructors and students in these courses. I provide assistance for everything from simple classroom and technology needs to advising on how to enhance courses with more experience-based curriculums. With the addition of several more courses offered at ARDEC, I also worked to develop and design an innovative teaching lecture hall in the Taylor Auditorium. This auction ring with pull out bleacher seating was transformed into a flexible seating classroom, which can accommodate up to 200 students sitting at tables in rows, groups, or small learning communities. Not only did I oversee the design and purchasing of the furniture to accommodate the flexible seating, I also designed the technology for the space. The flexible seating arrangement created a need to project information on more than just one screen. As a result, 6 TVs were purchased but linking them all on a small budget was a challenge. After research and trials, I was able to link all the technology in the room together and do it under budget. All technology can be linked, it can work independently or can be split in many configurations. However, though an innovative teaching classroom, the bleachers can still be pulled out and the space converted back to serve its original purpose as an auction ring/livestock exposition space as well. I have used my training as an agricultural educator and my expertise in designing and delivering experiential learning curriculum in agriculture to broaden and enhance student learning in the classes that I teach and in many other classes as well.

Professional Development Related to Teaching

Professional development related to my growth as an instructor is a continual and ongoing process. As a teacher educator, most of my teaching, outreach and research is focused on identifying and implementing “best practices” in the classroom. My involvement in the American Association for Agricultural Education (AAAE), the professional research organization for agricultural educators, is one example of my commitment to developing professionally. In AAAE, I am a member of the technical and mechanical agriculture Special Interest Group (SIG). At both regional and national meetings, members of the SIG group meet to discuss successes and challenges in our courses, new trends in technical agriculture, how to most effectively deliver content, and many other topics. My research is a means of professional development for me as well. I completed an Action Research study in my AGED 240 course in which I used data provided through a survey of current students to adjust my instruction within the semester. This data not only informed my instruction and altered how I taught, it led to the development of a teaching model used by other instructors at both the secondary and post-secondary levels. My involvement in the AAAE SIG, participation in research sessions at conferences, and conducting my own research has directly impacted my instruction both by advancing my technical agriculture curriculum as well as the pedagogical practices I implement.

Much of my service and outreach also contributes to my growth as an instructor, specifically in the area of technical and mechanical agriculture. My participation as a committee member (current associate superintendent) on the National FFA Agricultural Technology and Mechanical Systems competition is a prime example. I gain insights into new curriculum elements from others instruction of similar courses, which I implement into my courses. My curriculum, especially in the electrical systems area has improved drastically through interactions with other instructors focused in this area. To initiate professional development opportunities for secondary agriculture teachers in Colorado, I am frequently working with local industry partners to understand what specific skills they seek in our agricultural education graduates. These industry interactions guide my curriculum as well. Partnerships and my work with agriculture equipment dealers/manufacturers, Colorado commodity groups, and local producers have all informed and impacted my curriculum and instructional practice. I utilize these interactions to ensure my content is as current and industry relevant as possible.

Increasing content knowledge and pedagogy within the agricultural education community is not the only focus of my professional development. As a teacher educator, I also engage with faculty in the School of Education within the Center for Educator Preparation (CEP). Most recently, we in CEP engaged in several meetings/workshops regarding socially responsible teaching. Last year, as a faculty we completed a book study on the book *Rac(e)ing to Class: Confronting Poverty and Race in Schools and Classrooms*. During the study, we discussed addressing race and poverty with our students and how to incorporate these topics into teacher development programs. Currently, along with my CEP colleagues, we are participating in several workshops led by Fleurette (Flo) King from the Thriving Inclusion group. These workshops focus on how we as teacher educators can model inclusion and culturally responsible teaching in our classrooms. Additionally, we have tools to instruct our students on how to be more culturally responsible in their own classrooms. As an educator of current and future educators, I spend most of my time thinking, working, and teaching about how to “do it better”. This is a glimpse of a few areas where I feel I have grown the most in both my technical content as well as my pedagogy and classroom climate over the past few years.

CV SECTION 4: Evidence of Outreach/Service/Engagement

Service and outreach make up a significant portion of my appointment beyond instruction. My role as facilitator of experiential learning at ARDEC is only one aspect of my service appointment. In addition, I also manage the facilities and events held at the CoBank Center for Agricultural Education. This facility is not only a site for teaching and learning, it also hosts multiple meetings and conferences throughout the year. Various groups such as agriculture commodity organizations, agribusiness organizations, CAS stakeholders, education entities, etc. use the space for meetings and conferences regularly. Annually, the CoBank Center for Agricultural Education is reserved over 200 days for group meetings, accommodating over 4000 people utilizing the space. This is in addition to students occupying rooms while taking courses at ARDEC. These groups include commodity organizations, donors to the university, school district administrators, agriculture equipment dealers, agriculture educators, etc. I work with each group individually to ensure their experience with CSU is a positive and welcoming one. I have fostered many relationships with these groups, which have led to their deeper engagements with CSU in other meaningful ways. This increased support has been in the financial area as well as in kind engagements. I have also summarized below a list of other outreach and service engagements I have participated in while at CSU.

COMMITTEES

College of Agricultural Sciences Leadership Team, Non-Tenure Track Faculty Representative, 2023 – 2024

College of Agricultural Sciences, Non-Tenure Track Committee, CHAIR 2022 – 2024

College of Agricultural Sciences, Shepardson/Nutrien Building Classroom Design Committee, 2019 – 2020

College of Agricultural Sciences, Non-Tenure Track Committee, 2018 – present

College of Agricultural Sciences, Experiential Learning Committee, 2016 – present

College of Agricultural Sciences, Scholarship Selection Committee, 2015 - present

College of Agricultural Sciences, Academic Affairs Committee, 2012 - 2015

College of Agricultural Sciences, RamCamp, 2012 – present

PROFESSIONAL AFFILIATIONS AND ACTIVITIES

American Society for Farm Managers & Rural Appraisers Colorado Chapter, education board member 2022 - present

American Association for Agricultural Education, member 2012 - present

Agricultural Mechanics, AAAE SIG, member 2013 – present

Ag Ed Alliance (formerly Alpha Tau Alpha at Colorado State University), Ag Ed Student Organization, co-advisor, 2012 – present

Colorado TEAM AgEd (Joint Staff for Strategic Planning in Colorado Agricultural Education), member, 2012 – present

Career Pathways Collaborative Committee Member, Responsible for Creating National Assessments for Career and Tech Ed, 2011 – 2017

OTHER ACTIVITIES/ACCOMPLISHMENTS – SERVICE/OUTREACH

National FFA Agricultural Technology Career Development Event Associate Superintendent, Electricity Content Development Chair, 2023 – present

Listed as a contributing member on \$500,000.00 Inclusive Higher Ed Education Grant, funding the implementation of the Ram Scholars Program, 2023-present

Colorado FFA State Parliamentary Procedure Competition Leadership Event Superintendent, 2022 – present

Consultant in helping evaluate new courses in the Agribusiness and Food Innovation Management Master's degree program, 2022

Member of the implementation team for the Ram Scholars Program, designed to provide education and career training in agriculture for individuals with intellectual and developmental disabilities, 2022-present

Contributing member on \$147,000 Larimer County Economic & Workforce Development Workforce Innovation Grant, funding the implementation of the Ram Scholars Program, 2022-present

UpSkill Academy contributing member for implementation of the first “badging” curriculum rollout, 2022-present

Contributing member on \$256,723.00 CHS Foundation funded grant titled Student Driven Precision Tools for Sustaining Crop Water Resources and Soil Health, 2021-present

Awarded \$160,000.00 Federal Perkins Challenge Grant in partnership with Southwest Colorado BOCES and Colorado Community College System to deliver engine and equipment technical training for secondary agriculture teachers, 2020-2021

Colorado State FFA Agricultural Technology Career Development Event Superintendent (200+ students annually), 2013 - present

National FFA Agricultural Technology Career Development Event Committee Member and Volunteer, 2014 - 2023

Colorado State FFA Leadership Development Events, Co-Event Manager (230 students annually), 2013 - 2018

The National Council for Agricultural Education Agriculture, Food, Natural Resources Standards Revision Committee, Content Reviewer, 2015

Agriculture Teacher Professional Development Presentations:

2025 “Neurodiverse Learners in Ag Round 2” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Content Developer*, Fort Collins, CO

2025 “Laboratory Management” Colorado Ag Teachers Association Mid-Winter Conference,

Organizer, Content Developer, Presenter, Fort Collins, CO

2024 “Basic Wiring” Colorado Ag Teachers Association Summer Institute, *Presenter*, La Junta, CO

2024 “Ideas Unlimited Curriculum Sharing Presentations” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer*, Fort Collins CO

2024 “Neurodiverse Learners in Ag” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Content Developer, Presenter*, Fort Collins, CO

2023 “Equipment and Engine Training Council Small Gas Engine Instructor Workshop” Sponsored by EETC, Attendees from 6 different state, *Organizer, Content Consultant, Presenter*, Fort Collins, CO

2023 “CSU Ag Ed Program Connected Programming: How CSU Ag Ed is working to benefit the Ag Ed Profession” Colorado Ag Teachers Association Summer Institute, *Presenter*, Cedaredge, CO

2023 “Ideas Unlimited Curriculum Sharing Presentations” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer*, Fort Collins CO

2023 “Trailer Wiring and Fault Diagnosis/DC schematic and wiring” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Content Developer, Presenter*, Fort Collins, CO

2023 “AgNext’s Goal of Achieving Sustainability - Dr. Kim Stackhouse-Lawson, Director of AgNext and Professor at Colorado State University” Colorado Farm Show/Colorado Ag Teachers Association Mid-Winter Conference, *Organizer*, Greeley, CO

2023 “CAM’s on the Road: Bridging Agriculture Literacy Programming in YOUR Community - Jenny Bennett, Ag Literacy Outreach Coordinator, CAM’s Ag Academy” Colorado Farm Show/Colorado Ag Teachers Association Mid-Winter Conference, *Organizer*, Greeley, CO

2022 “UpSkilling Soil Moisture Sensor Badging Teacher Implementation Workshop” CHS Grant UpSkilling Academy Training Workshop, *Organizer, Presenter*, Fort Collins, CO

2022 “UpSkilling Irrigation Scheduler Badging Teacher Implementation Workshop” CHS Grant UpSkilling Academy Training Workshop, *Organizer, Presenter*, Fort Collins, CO

2022 “CSU UpSkilling Academy Opportunities for Teachers and Students Workshop” Colorado Ag Teachers Association Summer Institute, *Presenter*, Alamosa, CO

2022 “Ideas Unlimited Curriculum Sharing Presentations” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer*, Fort Collins CO

2022 “Trailer Brake and Wiring Tester Teacher Workshop” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Content Developer, Presenter*, Fort Collins, CO

2021 “Compact Engine Electrical and Hydraulic Systems Teacher Workshop” Perkins Challenge Grant Workshop in cooperation with Colorado Community College System and Southwest BOCES, *Organizer, Presenter*, Fort Collins, CO

2021 “Briggs and Stratton Small Gas Engine Teacher Workshop” Perkins Challenge Grant Workshop in cooperation with Colorado Community College System and Southwest BOCES, *Organizer, Presenter*, Fort Collins, CO

2021 “Building and Programming an Arduino Irrigation and Sensor System” Colorado Ag Teachers Association Virtual Mid-Winter Conference, *Organizer, Presenter*, Virtual from Fort Collins, CO

2020 “Soil Sensors and Irrigation” Colorado Ag Teachers Soil Sensor Professional Development, *Organizer, Presenter*, Rocky Ford, CO

2020 “Arduino Soil Sensors” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Presenter*, Fort Collins, CO

2019 “Mathematics and Hydraulics, the Application of Math through teaching Hydraulics,” Colorado Agriculture Teachers Association Summer Institute, *Presenter*, Steamboat Spring, CO

2019 “Agricultural Education Bootcamp-Maintenance and Management of the Ag Ed Laboratory” Workshop for new secondary agriculture teachers in Colorado, Fort Collins, CO

2019 “Welder Setup, Maintenance, and Operation Deep Dive Workshop” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Presenter*, Fort Collins, CO

2018 “Mountain States Agricultural Mechanics Instructors Conference” Sponsored and Coordinated by American Technical Publishers, *Organizer, Presenter*, Fort Collins, CO

2018 “Ag Mechanical and Technical Round Robin Workshop” Colorado Ag Teachers Association Summer Institute, *Organizer, Presenter*, Fort Collins, CO

2018 “Agricultural Education Bootcamp-Maintenance and Management of the Ag Ed Laboratory” Workshop for new secondary agriculture teachers in Colorado, Fort Collins, CO

2018 “Working with the Math Teacher, how to Reinforce Math in Ag” Colorado Ag Teachers Association Mid-Winter Conference, *Presented by Clark, N., & Clark J.*, Fort Collins, CO

2018 “Building and Troubleshooting the CAN Bus System in Tractors Deep Dive Workshop” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Presenter*, Fort Collins, CO

2018 “What is the CAN Bus System in Tractors?” Colorado Farm Show/Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Presenter*, Greeley, CO

2017 “Basics of Teaching Hydraulics,” Colorado Ag Teachers Association Summer Institute, *Organizer, Presenter*, Ignacio, CO

2017 “Trailer Wiring Mockups Deep Dive,” Colorado Ag Teachers Association Mid-Winter Conference, *Organizer, Presenter*, Fort Collins, CO

2016 “Teaching how to Install, Troubleshoot, and Maintain Landscape Irrigation Systems,” Colorado Ag Teachers Association Summer Institute, *Organizer, Presenter*, Cedaredge, CO

2015- Workshop, “Electrical Wiring and Plumbing in Agriculture,” Northeast BOCES Conference, Haxtun, CO

2015- Workshop, “Electrical Wiring and Plumbing in Agriculture,” Stratton Agriculture Teachers Conference, Stratton, CO

2015 “Utilizing the Agricultural Mechanics Psychomotor Development Model – A Training.” Presented at the Colorado Vocational Agricultural Teachers Association Mid-Winter Conference. *Presented by* Schlabach, K., Enns, K., & **Clark, N.**, Colorado Springs, CO

2013 - Workshop, WDA Dairy Curriculum. Colorado Vocational Agriculture Teachers Association Summer Conference. *Presented by* **Clark, N.**, Ralston, M., & Enns, K., Fort Collins, CO